

Reproduction of state-spaces/mamba: Mamba-370M zero-shot evaluation

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Independent reproduction · Artifact evaluation · Natural language processing

Abstract

Ran the README zero-shot evaluation for the released state-spaces/mamba-370m checkpoint with lm-evaluation-harness and the mamba_ssm backend, on the full LAMBADA, HellaSwag, PIQA, ARC-Easy, ARC-Challenge and WinoGrande sets, batch size 64. mamba_ssm and causal-conv1d were compiled at the pinned commits for sm_120 with CUDA 12.8 (MAMBA_KEEP_CUDA_BUILD=TRUE for the Mamba-1 kernels); apache-tvm-ffi was pinned to 0.1.9 so the bundled tilelang imports. One NVIDIA RTX 5060 Ti.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	Mamba-370M zero-shot, 6 tasks	2026-10-10	1	1	Signed by the K-Veritas server

2 Results

r1: Mamba-370M zero-shot, 6 tasks

Run	Metric	Final value	Step
1	lambada_openai_sample_len	5153	
1	lambada_openai_perplexity	8.13741	
1	lambada_openai_acc	0.556375	
1	hellaswag_sample_len	10042	
1	hellaswag_acc	0.372037	
1	hellaswag_acc_norm	0.464848	
1	piqa_sample_len	1838	
1	piqa_acc	0.695321	
1	piqa_acc_norm	0.683351	
1	arc_easy_sample_len	2376	
1	arc_easy_acc	0.549242	

Run	Metric	Final value	Step
1	arc_easy_acc_norm	0.481902	
1	arc_challenge_sample_len	1172	
1	arc_challenge_acc	0.25	
1	arc_challenge_acc_norm	0.277304	
1	winogrande_sample_len	1267	
1	winogrande_acc	0.555643	

Run	Inline claim	Value	Phase
1	lambada_openai_perplexity	8.13741	
1	lambada_openai_acc	0.556375	
1	hellaswag_acc	0.372037	
1	hellaswag_acc_norm	0.464848	
1	piqa_acc	0.695321	
1	piqa_acc_norm	0.683351	
1	arc_easy_acc	0.549242	
1	arc_easy_acc_norm	0.481902	
1	arc_challenge_acc	0.25	
1	arc_challenge_acc_norm	0.277304	
1	winogrande_acc	0.555643	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	316.0 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	7c7e8faf5459ef60b4e88562d423721f85fa44a22c9ede7b9a239d924cc5f0cb	2.9 MB
r1.kvbundle.zip	7b483483fd2b5bfb57293081c8076c98b1aa147020fc7bae55c1a1be7f43cfc6	279 KB

5 Links and licence

Paper: <https://arxiv.org/abs/2312.00752>

Reproduction of: <https://github.com/state-spaces/mamba/tree/e9594ce1c732d97440f0332fdc43170a2294dbfa>

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: mamba, state-space-models, zero-shot, lm-evaluation-harness.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

- [1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.
- [2] Sealed experiment report r1 (Mamba-370M zero-shot, 6 tasks), kv:2610.00033v1/r1. Data hash 1ca0f6d826b86b3262aedbf1866626edd8c23b0e04c825c7febf92fd61ead5e2.
<https://kveritas.org/records/2610.00033v1/r1.pdf>